Class Test II
CITS2200 Data Structures and Algorithms

May 9, 2024

- Total marks for this paper is 10. Please answer by filling the appropriate circle for each question in your answer sheet.
- You should carefully fill your student number in the answer sheet.
- **Please return only the answer sheet, do not return the question paper. Take it away with you and throw it in a recycling bin.**

Q1. A cyclist can travel 24 km/hour when wind is blowing from behind, and 12 km/hour against the wind. The cyclist travels A to B and then B to A. The wind is blowing from A to B throughout the travel of the cyclist. What is the amortized speed of the cyclist?

(b)

Assume the distance between A and B is x Km. Simple

(a) 12 km/hour; **(b)** 16 km/hour; **(c)** 20 km/hour; **(d)** Impossible to say without knowing the distance from A to B.

Q2. Each node of a doubly-linked list can store an `item` and two references `next` and `previous` to the nodes to the right and left of the node respectively. There are three nodes A, B and C in a doubly-linked list, such that `A.next=B` and `B.next=C`; `C.previous=B` and `B.previous=A`. We have used the names of the nodes as their references. We want to delete the node with reference B. Which one of the following is a correct way to delete B from the doubly-linked list?

C.previous.previous is A.
A.next = C; disconnects B

(b)

(a)
```
A.next=B.next
C.previous=A.next
```

(b)
```
C.previous.previous.next=C
C.previous=C.previous.previous
```

C.previous = A

(c)
```
C.previous.next=C
C.previous=C.previous.previous
```

(d)
```
A.next=B.next
C.previous=B.previous.next
```

Q3. Each node of a singly-linked list can store an `item` and a reference `next` to the node to its right. There are two nodes A and C in a linked-list such that `A.next=C`. You have created another node whose reference is B. Which of the following connects B correctly between A and C?

(a) `A.next=B`
`B.Next=C`

(b) `A.next=C`
`B.next=C`

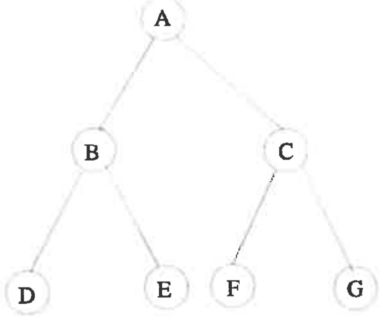
(c) `A.next=B`
`B.next=A`

(d) `B.next=C`
`A.next=C`

(a)

Q4. Which is the correct `inorder` enumeration of the nodes of the following binary tree if you start traversal from the root node A

this is not included



(d)

(a) $A-B-C-D-E-F-G$; **(b)** $A-B-D-C-E-F-G$; **(c)** $B-A-D-E-C-F-G$; **(d)** $D-B-E-A-F-C-G$.

Q5. You have an input array storing the items A, B, C, D, E from index 0 to index 4. You first enqueue these items from index 0 to index 4. Now you dequeue the items one by one and push them into a stack. Finally, you pop the elements from the stack one by one. The order of the popped items is:

(d)

E, D, C, B, A in the correct order

(a) E, D, C, A, B; **(b)** A, B, C, D, E; **(c)** B, A, C, D, E; **(d)** none of the above.

Q6. You are storing integers $26, 17, 14, 8, 5$ in an array of size 5, using a hash function $N \ mod \ 5$ when N is the integer you are storing. Which of the following is a correct representation of the array after storing these integers?

divide each number by 5 and the remainder is the index

(a)

17	26	14	8	5

(b)

26	17	14	8	5

(c)

5	26	17	8	14

(d)

5	17	14	8	26

(c)

Q7. A `max-heap` is stored in an array as a binary tree, with the highest priority item at the root. The array location with index 0 is empty, and all other array locations store the priorities of the elements in the `max-heap` in the usual way of storing a binary tree using an array. Which of the following arrays correctly represents a `max-heap`?

This is not included

(a)		45	13	24			34	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(b)		45	13	7			24	34					13	3						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(c)		45	13	34			24	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(d)		45	24	13			34	21					7	3						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c)

This question is not included

Q8. The highest priority item is now deleted from the `max-heap` and an item with priority 16 is inserted. Which of the following correctly represents the new heap after rearrangement?

(a)		45	13	24			34	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(b)		16	13	34			24	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(c)		34		13	24		16	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
(d)		34	13	24			16	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(d)

Q9. Which of the following statement is false for a singly-linked list? Assume that the `next` references go from left to right, and you are given a reference called `current` of a node in the middle of the linked list and there are n nodes to the left and right of `current`. You are also given the reference of the `first` node of the linked list.

(c)

(a) The complexity of finding the reference of a node 5 nodes to the right of `current` is $O(1)$;
(b) The complexity of adding a node to the right of `current` is $O(1)$;
(c) The complexity of adding a node to the left of `current` is $O(1)$; → requires traversing from first
(d) The complexity of counting the number of nodes in the linked list is $O(n)$.

Q10. Which of the following statement is true for a doubly-linked list? Assume that you are given a reference called `current` of a node in the middle of the linked list and there are n nodes to the left and right of `current`. You are also given the reference of the `first` node of the linked list.

(a) The complexity of counting the number of nodes in the doubly-linked

list is $O(\log n)$;
(b) The complexity of adding a node to the left or right of `current` is $O(1)$;
(c) The complexity of counting the number of nodes in the doubly-linked list is $O(1)$.
(d) The reference of the last node in the doubly-linked list can be found in $O(1)$ time.

(b)

———End of Paper———